Overall Goals:

- Design a rainwater catchment system to supplement the community's existing water supply, taking into account local weather patterns and materials availability.
- Develop a storage system for collected rainwater, including considerations for capacity, durability, and ease of use.
- Integrate a filtration or treatment system to ensure the collected water is safe to drink.
- Create an implementation plan for constructing and installing the rainwater catchment and storage system.

Milestones:

January
- Intro and engagement

February
- Research rainwater catchment designs, including gutter systems, collection tanks, and filtration/treatment methods.
- Study local rainfall patterns and estimate the potential water yield based on the community's needs and available roof sizes.
- Begin conceptual designs for the rainwater catchment and storage system.

March
- Sketches of design
- Continue working on planning filtration/treatment methods
- Beginning of prototyping (low-fidelity prototype)

April
- Working prototype + future design goals and improvements
- End of Semester Showcase!

May
-

SMART Weekly Goals:

Ex: Week 1

- Start a research document about water catchment designs.
    - Have a conversation discussing everyone's design experience and start a rough brainstorm ways we can prototype
    - Compile a list of at least 3 different designs.
    - Identify how they differ, which situations they are most applicable in, what information is needed to decide on one vs another system.
    - Make a list of the information you need to know about the community (elevation, rainfall patterns, etc) in order to decide on which system to use.
    - Come up with some follow-up questions to investigate in Week 2

Week 2

- Study current approaches to catchment and water storage to inform our conceptual designs
    - Incorporate previous research and narrow down brainstorming

        - Have all research organized and easy to access
    - Finish week 1 goals